

FROM TRANSPARENT TO OPAQUE: LEVERAGING GATING MECHANISM IN LATENT REASONING CHAIN

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ABSTRACT

We proposed a novel paradigm for implementing latent reasoning by combining DAPO-like reinforcement learning and special gating units. In this way, we transformed text-based explicit reasoning models to latent reasoning models, progressively and practically. Initial experiments demonstrate its feasibility, with significant stability and the least extra training overhead by parallelism and minimal changes, while holding full potential to scale up.

1 INTRODUCTION

The Chain of Continuous Thought (COCONUT) (Hao et al., 2024) explores the potential of the latent layer of reasoning as an alternative to the traditional explicit CoT (Wei et al., 2022) in both the reasoning procedure and the response procedure. Its previous work has shown that the latent layer of continuous reasoning can reduce unnecessary statements and focus on the core of the problem.

These models are issued from general-purpose language, trained in supervised steps of a reasoning procedure, and progressively increase their layers of latent reasoning. It is impractical due to the lack of customized reasoning data, and progressive training is significantly less convenient than explicit CoT. Hence, we provide a new approach to implementing this latent reasoning layer, fine-tuning from explicit reasoning models.

In 2014 (Chung et al., 2014), Chung et al. proposed the Gated Recurrent Unit. It is first leveraged to improve the RNNs, but we found that using it in the latent reasoning chain is also feasible to maintain its long-term contextual awareness.

In January, the DeepSeek team proposed a distillation-based method to implement a reasoning model through reinforcement learning (DeepSeek-AI, 2025), and applied the approach to Qwen-2.5B (Yang et al., 2024) to a reasoning model. Given the limitations of a 2.5B parameter model in reasoning tasks, we aim to implement COCONUT on the small model and find its potential in reasoning compared to the original explicit reasoning model.

2 BACKGROUND

Explicit Reasoning Models Advances in explicit reasoning models, including OpenAI o1 & o3 (OpenAI, 2025; 2024) and DeepSeek R1 (DeepSeek-AI, 2025), demonstrates its strong ability on mathematical reasoning. Based on Chain-of-Thought (CoT, Wei et al., 2022), models can answer demanding questions after solving it internally with explicit procedures.

However, explicit reasoning may have difficulty in concision and efficiency—explicit reasoning text is often verbose and inefficient.

Chain of Continuous Thought Hao et al. proposed COCONUT (2024), assigning models to “reason in a latent continuous thought.”

Compared to explicit reasoning, COCONUT leverages latent spaces, promoting its efficiency and accuracy. However, it faces difficulty in scaling up, and requires step-by-step guidance in training.

Large-scale Reinforcement Learning Large-scale Reinforcement Learning (RL), including Proximal Policy Optimization (PPO, Schulman et al., 2017), Group Relative Policy Optimization (GRPO, Shao et al., 2024), and Decoupled Clip and Dynamic Sampling Policy Optimization (DAPO, Yu et al., 2025).

We adapt the DAPO objective, which is formulated as:

$$\begin{aligned} \mathcal{J}_{\text{DAPO}}(\theta) = & \mathbb{E}_{(q,a) \sim \mathcal{D}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|q)} \\ & \left[\frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \right] \\ \text{subject to } & 0 < |\{o_i \mid \text{is_equivalent}(a, o_i)\}| < G, \end{aligned} \quad (1)$$

where

$$r_{i,t}(\theta) = \frac{\pi_{\theta}(o_{i,t} \mid q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t} \mid q, o_{i,<t})}, \quad \hat{A}_{i,t} = \frac{R_i - \text{mean}(\{R_i\}_{i=1}^G)}{\text{std}(\{R_i\}_{i=1}^G)}. \quad (2)$$

3 METHODOLOGY

3.1 MODEL ARCHITECTURE

Modern LLMs involves Embedding, Transformer, and Output layers, as we denote them $\mathbf{E}$, $\mathbf{T}_n$, and $\mathbf{O}$ respectively, where n indicates the layer of hidden state.

To avoid breaking special usage in final layers (Elhage et al., 2021), we used middle layers as the entry of the latent reasoning space.

3.2 INFERENCE

The whole inference procedure is described in (...)

After extracting the hidden state $\mathbf{h}_{n,0}$ (where n denotes the layer number and k denotes time sequence, which is 0 here), we pass the state to the Variational Autoencoder (Kingma and Welling, 2019), and produce a vector for storage with length l_s .

Then, the new vector with the same shape of $\mathbf{h}_{n,k}$, as we denote $\mathbf{h}'_{n,k}$, will be operated by several gates, as described in Equation 3.

$$\mathbf{h}_{n,k} = \mathbf{h}_{n,k-1} \odot \mathbf{g}_k + (1 - \mathbf{g}) \odot \mathbf{t}_k \quad (3)$$

Gates There are three gates in our model, which are the general gate, the injection scalar, and the Dynamic Current Step Mixing Gate.

1. **General Gate.** The general gate is derived from sequence, its value dynamic. We denote it as $\mathbf{g}_t$, and its procedure of derivation can be described in Equation (not yet).
2. **Injection Scalar.** The injection scalar is a constant value, which brought the result of gate-shaper to the gate, to bridge the difference in order of magnitude.
3. **Dynamic Current Step Mixing Gate.**

Long-term Cache To enable long-term storage of reasoning features, we introduce a VAE-style bottleneck layer, compressing 1536-dimensional features to 256 dimensions and then expanding back to 1536 before gating.

We store it in each operation, and concatenate these vectors into a matrix subsequently as hidden state.

3.3 TRAINING

We use the DAPO objective as our reinforcement learning framework, but the whole procedure is slightly different from the original DAPO.

3.3.1 REINFORCEMENT LEARNING

The whole procedure of reinforcement learning can be described in Figure.

Firstly we apply rigorous datasets to the inference procedure, and verify the answer (Kydliček).

We parallelized several inference procedures simultaneously, preferably 16 for a same question, and verify the answer.

Award Mechanism We use $\text{award}_{\text{acc}}$ and $\text{penalty}_{\text{acc}}$ to award or penalize in accuracy. After mixing the accuracy and length evaluation, we inject it to loss.

Positive-Negative Ratio We use a ratio to mix positive and negative cases, which is 1 : 1.

3.4 TIME CONSISTENCY REGULARIZATION

A significant reason for chaotic response of multi-layer COCONUT is its time inconsistency, which lead to the model’s failure to maintain a consistent reasoning process.

We adapt a mechanism called “Gating Value Bonus,” which is used to guarantee the gate to converge to a fixed position, as we denote in Equation 4.

$$\mathcal{J}_{\text{gate}} = \min \left\{ C_0 \exp \left[\lambda \left(\frac{1}{2} - g_i \right)^2 \right] \right\} \quad (4)$$

3.5 PARALLELIZED TRAINING

After sampling, we concatenate the question, answer, and hidden state, and apply it to training procedure.

4 EXPERIMENTS

5 DISCUSSION

6 CONCLUSION

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A APPENDIX